CompTIA A+
Hardware Components & System Assembly
AP-A1 Complete Assessment Assignment

Section A - Concept Application

S1 - Selecting Compatible Core Hardware

Scenario: You are setting up a new office workstation and need to choose the core hardware for it from a supplier's parts list.

Question: Explain the process of selecting a compatible motherboard, CPU, RAM, and storage device for a new office workstation, and justify how you would confirm they will work together.

Answer

A reliable build starts by choosing components in a logical order and checking every electrical, physical, and firmware dependency before purchase. The motherboard is the central compatibility point because the CPU, RAM, storage and expansion devices all connect to it.

1. Define the office workload.

For a normal office workstation, the expected workload includes web browsing, email, word processing, spreadsheets, video meetings and light multitasking. This generally does not require a high-end CPU or gaming-class GPU, but it benefits from sufficient RAM and fast solid-state storage.

2. Select the CPU.

Choose a modern mid-range processor that meets performance and budget requirements. Important CPU specifications include core/thread count, socket type, thermal design power, integrated graphics availability, and supported memory generation. A processor with integrated graphics can reduce cost when the office does not need a discrete GPU.

3. Match the motherboard to the CPU.

The CPU socket on the motherboard must physically and electrically match the CPU. For example, an Intel LGA1700 CPU requires an LGA1700 motherboard, while an AMD AM5 processor requires an AM5 motherboard. The chipset must also support the processor generation. A matching socket alone is not always enough because an older BIOS may not recognize a newer CPU.

4. Select compatible RAM.

Check the motherboard manual for the supported memory generation, such as DDR4 or DDR5, maximum capacity, supported speed, number of DIMM slots and recommended channel configuration. DDR4 and DDR5 are physically keyed differently and cannot be interchanged. For an office workstation, 16 GB is normally a strong baseline because it supports multitasking without excessive paging to storage.

5. Select storage.

Check whether the motherboard provides SATA ports, M.2 slots, or both. A 2.5-inch SATA SSD requires a SATA data connection to the motherboard and SATA power from the PSU. An M.2 NVMe SSD installs

directly into an M.2 slot and communicates through PCI Express. The M.2 slot must support the selected drive type and length, such as M.2 2280.

6. Confirm physical and power compatibility.

Confirm that the motherboard form factor fits the case, that the CPU cooler fits the socket and case height, and that the PSU provides the 24-pin ATX motherboard connector plus the required CPU EPS connector. If a discrete GPU is added, check slot clearance and GPU power requirements as well.

Compatibility confirmation method:

- Compare the CPU model with the motherboard manufacturer's official CPU support list.
- Check the minimum BIOS version required for that CPU.
- Check the motherboard memory support page or Qualified Vendor List (QVL) for RAM type, capacity and supported speed.
- Check the motherboard storage specification for SATA and M.2/NVMe support.
- Confirm case form factor, PSU connectors and power capacity.
- After assembly, enter BIOS/UEFI and verify that CPU, total RAM and storage are all detected correctly.

Justification: These checks prevent common failures such as a system that does not POST, RAM that operates incorrectly, an NVMe drive that is not detected, or a CPU that requires a BIOS update before the computer can start.

S2 - HDD, SATA SSD and NVMe Storage

Scenario: You are advising a client whose employees mainly run office and productivity applications on their workstations.

Question: Explain the differences between HDD, SSD, and NVMe storage, and recommend the most suitable option for this workload, giving technical reasons for your choice.

Answer

Feature	HDD	SATA SSD	NVMe SSD
Technology	Magnetic rotating platters and moving read/write heads	NAND flash memory; no moving parts	NAND flash memory using PCI Express and NVMe protocol
Typical interface	SATA	SATA	PCIe through M.2 or add-in card
Relative speed	Slowest	Much faster than HDD	Fastest of the three
Latency	High because of mechanical movement	Low	Very low
Noise	Audible vibration/rotation	Silent	Silent
Shock resistance	Lower	Higher	Higher
Cost per GB	Lowest	Moderate	Moderate to higher depending on model
Best use	Bulk/archive storage	General-purpose desktop storage	High-performance system and application drive

Recommended option: A 500 GB to 1 TB NVMe SSD, provided the motherboard has a suitable M.2 NVMe slot. If budget is very limited, a SATA SSD is still a strong alternative and should be preferred over an HDD for the operating-system drive.

Technical justification: Office applications generate many small file operations during boot, login, application launch, updates, browser caching and document access. Low storage latency has a strong effect on perceived responsiveness. NVMe uses PCI Express rather than the older SATA interface,

allowing higher throughput and more efficient command handling. Although office workloads rarely use the maximum sequential speed of a premium NVMe drive, the low latency and fast random access make the workstation feel more responsive. In addition, SSDs have no moving parts, generate less noise, resist shock better and normally use less power than desktop HDDs.

An HDD is still useful when very large, low-cost local storage is required, but it is not the best primary drive for a modern office workstation because boot time, application loading and updates are noticeably slower.

S3 - GPU and Cooling Under Sustained Load

Scenario: You are configuring a workstation that will run mildly graphics-intensive design software for several hours a day.

Question: Explain the role of the GPU and the cooling system in maintaining performance and stability under this kind of sustained load.

Answer

The GPU is specialized for highly parallel graphical calculations. Design applications can use it to render 2D/3D views, accelerate effects, manipulate high-resolution images, display complex models and in some cases perform compute operations. A suitable GPU improves interactive responsiveness and can reduce the time required for supported rendering tasks.

Sustained graphical work creates continuous electrical load and heat. Modern CPUs and GPUs monitor temperature and power conditions. If cooling is insufficient, they may reduce clock speed automatically; this is called thermal throttling. Throttling protects the hardware but reduces performance. Severe cooling problems can also cause application crashes, display-driver resets, unexpected shutdowns or reduced component life.

- The GPU heatsink transfers heat away from the GPU die.
- GPU fans move air across the heatsink fins.
- Thermal interface material improves heat transfer between the chip and cooler.
- Case intake and exhaust fans replace hot internal air with cooler external air.
- The CPU cooler prevents CPU overheating while the design application is also using CPU resources.
- Good cable management prevents cables from blocking airflow.

For stable operation, I would confirm that the GPU meets the software vendor's requirements, that the PSU can supply the required power, that case airflow is unobstructed and that temperatures remain within manufacturer limits during a sustained workload. I would also keep heatsinks and filters free of dust and verify that all fans operate correctly.

S4 - Importance of Hardware Compatibility

Scenario: You are assembling a PC from components sourced from three different vendors.

Question: Explain why hardware compatibility matters when assembling a computer, and identify potential issues that could arise from incompatible components.

Answer

Computer components are built around defined mechanical, electrical and communication standards. Parts from different manufacturers can work perfectly together when they follow the same compatible standards. Problems occur when one or more requirements do not match.

Compatibility Area	Possible Problem	Likely Result
CPU socket/chipset	CPU not supported by motherboard	CPU cannot be installed or system will not POST

BIOS/UEFI support	CPU requires newer firmware	Fans may spin but there may be no display or POST
RAM generation	DDR4 RAM selected for DDR5 board	Module physically cannot fit
RAM capacity/speed	Unsupported module density or profile	Instability, lower speed, or failure to boot
Storage interface	SATA-only M.2 drive used in NVMe-only slot or vice versa	Drive not detected
PSU wattage	Power supply is undersized	Crashes, shutdowns, instability
Power connectors	Required CPU/GPU connector missing	Component receives no power
Case/form factor	ATX board in incompatible case	Board does not fit or mounting holes do not align
GPU dimensions	Card too long/thick for case	Physical installation impossible
CPU cooler	Wrong mounting kit or insufficient clearance	Cannot install or CPU overheats

Compatibility therefore matters before purchase, during assembly and during first boot. The best practice is to check official manufacturer documentation instead of assuming that components will work together merely because their connectors appear similar.

S5 - PSU Selection, Safety and Connection Checks

Scenario: You are finalizing the build and need to connect the power supply unit (PSU) to the motherboard, GPU, and storage devices before first boot.

Question: Explain how you would choose an appropriately rated PSU for this build and describe the safety and connection checks you would carry out before powering it on.

Answer

The PSU must provide enough continuous power for the entire system and must also provide the correct connectors. I would estimate the maximum power draw of the CPU, GPU, motherboard, storage, RAM, cooling fans and attached devices, then choose a reputable PSU with additional headroom. Headroom avoids operating the PSU continuously near its limit and leaves capacity for transient loads and small future upgrades.

Example: If the estimated peak system requirement is about 400 W, a quality 550-650 W PSU is a reasonable range for many builds. The exact selection should follow the CPU/GPU vendor recommendations. An 80 PLUS efficiency rating is useful, but build quality, protection features and correct connectors are equally important.

- 24-pin ATX connector for the motherboard.
- 4-pin or 8-pin EPS/CPU power connector near the CPU socket.
- PCIe 6-pin, 8-pin, or newer GPU power connector if the graphics card requires external power.
- SATA power connectors for 2.5-inch SSDs, 3.5-inch HDDs and some accessories.
- Sufficient cable reach for the selected case.

Safety and pre-power checks:

1. Switch the PSU off and disconnect the AC mains cable before making internal connections.

2. Use ESD protection: work on a non-carpeted surface where possible, discharge static electricity, and use an antistatic wrist strap connected correctly to ground/chassis according to lab procedure.

3. Confirm the motherboard is installed on the correct standoffs so the underside cannot short against the case.

4. Confirm the CPU cooler is firmly mounted and its fan is connected to the CPU_FAN header.

5. Connect the 24-pin ATX and CPU EPS power connectors fully; do not confuse CPU EPS and PCIe GPU connectors.

6. Connect GPU power if required and make sure the card is fully seated in the PCIe slot.

7. Connect SATA power and SATA data to SATA drives; an M.2 NVMe drive normally needs no separate power cable.

8. Check that no loose screws, tools or metal objects remain in the chassis.

9. Check cable routing so no wire can contact a fan blade.

10. Perform a final visual inspection, then reconnect AC power and switch on the PSU for first boot.

S6 - POST Verification and Troubleshooting

Scenario: You are handing over a freshly assembled workstation to the client and need to confirm it is ready for use.

Question: Explain the checks you would perform to confirm the system passes POST successfully, and what you would do if it did not.

Answer

POST (Power-On Self-Test) is the firmware-controlled hardware initialization performed when the computer starts. A successful POST shows that the essential hardware needed to begin the boot process is operating well enough for BIOS/UEFI to continue.

1. Press the power button and verify that the system powers on normally.

2. Check that CPU and case fans start and that there are no abnormal sounds or burning smells.

3. Observe the monitor for the manufacturer logo, POST information or BIOS/UEFI screen.

4. Check motherboard diagnostic LEDs, seven-segment codes or beep codes if the board provides them.

5. Enter BIOS/UEFI and confirm that the correct CPU model, expected total RAM and storage devices are detected.

6. Check CPU temperature in BIOS to make sure cooling is functioning normally.

7. Confirm that the correct boot drive appears in the boot list.

8. Allow the system to boot into the operating system and check Device Manager/system information for expected hardware.

If POST fails, I would troubleshoot from the simplest and most fundamental causes first:

1. Disconnect AC power before touching internal components.

2. Recheck the PSU switch, AC connection, 24-pin motherboard power and CPU EPS power.

3. Check front-panel power-switch wiring if the system does not react at all.

4. Reseat RAM and test one known-good module in the motherboard-recommended slot.

5. Reseat the GPU and verify GPU power; if the CPU has integrated graphics, test the motherboard display output where appropriate.

6. Check CPU seating, socket pins/contacts and cooler mounting if earlier checks do not solve the problem.

7. Disconnect nonessential drives and peripherals and attempt a minimal boot with motherboard, CPU, cooler, one RAM module and graphics output.

8. Use motherboard diagnostic LEDs/beep codes to identify the failing stage.

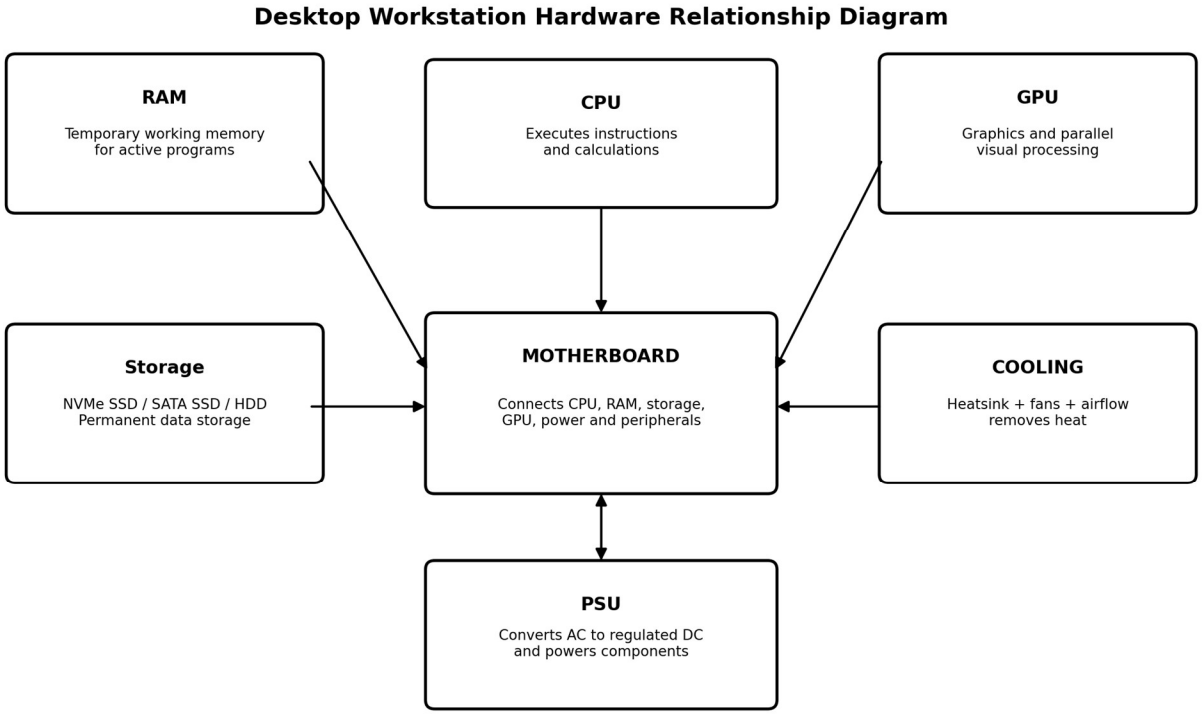
9. Clear CMOS if appropriate and verify BIOS compatibility with the CPU.

10. Swap in known-good compatible parts one at a time to isolate the faulty component.

Section B - Practical Hardware Tasks

Task 1 - Component Identification & Functions

Objective: Identify and document the motherboard, CPU, RAM, GPU, storage device and cooling system, including their function and visible specification.



Component	Example Make/Model or Key Specification	Function
Motherboard	B660-class micro-ATX motherboard, LGA1700, DDR4, M.2 NVMe, SATA	Main circuit board. Connects and allows communication between CPU, RAM, storage, GPU, PSU and peripherals. Provides chipset, firmware, slots, ports and power distribution.
CPU	Intel Core i5-12400, 6 cores / 12 threads, LGA1700	Executes program instructions, performs calculations, controls logical operations and coordinates system tasks.
RAM	16 GB DDR4, 2 x 8 GB DIMMs	Temporary high-speed working memory used by the operating system and active programs. Contents are volatile and are lost when power is removed.
GPU	Integrated Intel UHD Graphics or entry discrete GPU such as GTX 1650	Processes graphics and parallel visual workloads and produces display output. A discrete GPU provides greater graphics performance.
Storage	1 TB M.2 2280 NVMe SSD	Non-volatile storage for Windows, applications and user files. Retains data after power is removed.
Cooling system	Tower/top-down CPU heatsink and fan plus case intake/exhaust fans	Transfers and removes heat to keep CPU, GPU and internal components within safe operating temperatures.

Identification method: I would read the model number printed on the motherboard PCB, CPU label or BIOS information, RAM module sticker, GPU label, drive label and cooler/fan markings. I would then compare those markings with manufacturer documentation to confirm the specification.

Task 2 - Motherboard Ports & Socket Identification

Objective: Identify the motherboard physical layout, major ports, expansion slots and CPU socket type.

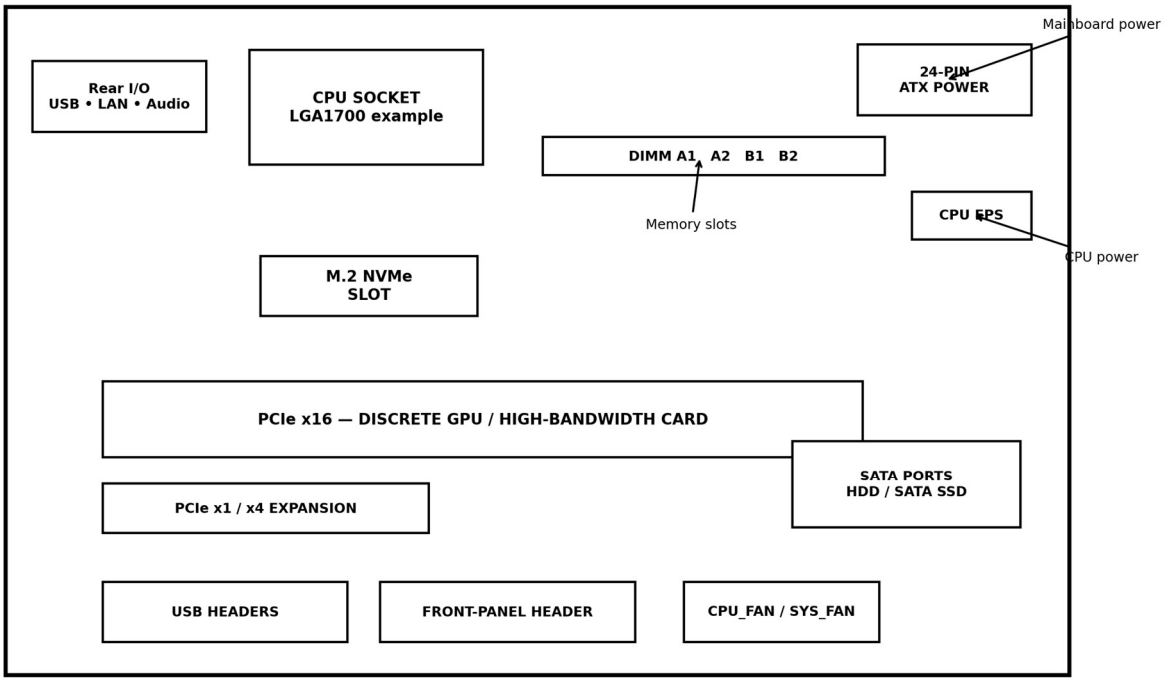
Motherboard Feature	Location / Identification	Purpose
24-pin ATX power	Large connector on board edge	Main motherboard power from PSU
8-pin CPU EPS power	Near top-left/top edge of CPU socket	Dedicated CPU voltage-regulator power
CPU socket	Center-upper motherboard area; example LGA1700	Holds CPU and provides electrical contacts
DIMM slots	Long slots beside CPU socket	Install DDR memory modules; matched slots support dual-channel operation
PCIe x16 slot	Long expansion slot	Typically used for a discrete GPU
PCIe x1/x4 slots	Shorter expansion slots	Network, sound, capture and other expansion cards
M.2 slot	Flat connector with mounting screw/standoff	Installs M.2 NVMe or supported M.2 SATA drive
SATA ports	Small keyed right-angle/vertical connectors	Connect SATA SSDs, HDDs and optical drives
USB 2.0 header	Internal pin header	Connects front-panel/internal USB 2.0 ports/devices
USB 3.x front header	Larger internal header	Connects high-speed front-panel USB ports
Front-panel header	Small pin block at board edge	Connects case power switch, reset switch and LEDs
CPU_FAN header	Near CPU socket	Powers and monitors CPU fan
SYS_FAN headers	Various board locations	Power/control case fans
Rear I/O ports	Rear edge of motherboard	External USB, network, audio, video and other ports

CPU socket example and family compatibility:

For the representative workstation, the socket is Intel LGA1700. LGA1700 motherboards can support selected Intel 12th, 13th and 14th generation Core processors depending on the exact chipset and BIOS version. The motherboard manufacturer's official CPU support list must be checked for the exact model and required BIOS version.

Text diagram - typical motherboard layout (not to scale):

Labeled Motherboard Layout — Typical ATX / micro-ATX Arrangement




```
+-------------------------------------------------------------+
| Rear I/O      CPU EPS                                       |
| [USB][LAN]    [8-pin]                                       |
|                 +-----------+     DIMM A1  DIMM A2          |
|                 | CPU       |     DIMM B1  DIMM B2          |
|                 | Socket    |                               |
|                 +-----------+        [24-pin ATX]           |
|                                                             |
|   [M.2 NVMe slot]                                           |
|                                                             |
|   [=========== PCIe x16 ===========]                        |
|   [==== PCIe x1 ====]                                       |
|                                          [SATA][SATA]       |
|   USB headers       Front-panel header       Fan headers    |
+-------------------------------------------------------------+
```

For the actual practical submission, a photograph of the lab motherboard can be inserted here and arrows can be added for the same labels. This text diagram provides the required labeled reference when a photo is unavailable.

Task 3 - RAM & Storage Installation

Objective: Safely remove and reinstall one RAM module and one storage device, then verify that both are detected.

A. ESD and Safety Preparation

1. Shut down the computer correctly.

2. Switch the PSU off and unplug the AC power cable.

3. Press the case power button briefly after unplugging to help discharge residual power.

4. Use an antistatic wrist strap according to lab procedure and avoid touching component contacts.

5. Handle RAM and storage devices by their edges.

B. RAM Removal and Reinstallation

1. Locate the DIMM slot containing the selected RAM module.

2. Open the retaining latch or latches at the ends of the slot.

3. Lift the RAM module straight out while holding only its edges.

4. Inspect the slot and module for dust, debris or obvious damage.

5. Align the notch in the RAM module with the keyed ridge in the DIMM slot. The module fits in only one orientation.

6. Press down evenly until the module is fully seated and the retaining latch or latches lock into place.

7. Visually check that both ends of the module are at the same height and fully inserted.

C. Storage Removal and Reinstallation - NVMe Example

1. Locate the M.2 NVMe drive on the motherboard.

2. Remove the retaining screw or tool-less latch while supporting the drive.

3. Allow the drive to rise slightly, then slide it gently out of the M.2 connector.

4. Inspect the connector and drive contacts for debris or damage.

5. Insert the NVMe drive at a slight angle into the M.2 slot until fully engaged.

6. Press the drive down to the standoff and secure it with the correct screw/latch. Do not overtighten.

7. Refit any M.2 heatsink or thermal pad in the correct position if provided.

If the practical system uses a SATA drive instead, disconnect and reconnect both the SATA data cable and SATA power cable, ensuring the keyed connectors are fully seated.

D. Detection Verification

1. Reconnect AC power and switch on the system.

2. Enter BIOS/UEFI.

3. Confirm that the expected total RAM capacity is shown. For two 8 GB modules, the expected total is approximately 16 GB.

4. Confirm that the NVMe/SATA storage device appears in the storage or boot-device information.

5. Boot into the operating system and check system information for RAM capacity and Disk Management/Device Manager for the storage device.

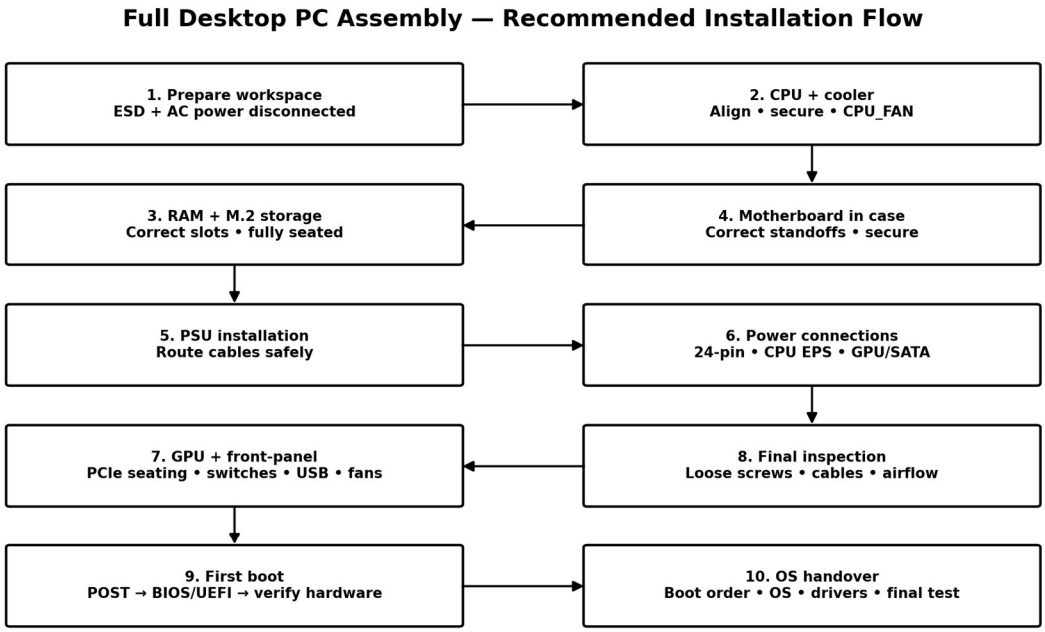
6. If either component is missing, power off and reseat it before further troubleshooting.

Verification Item	Expected Result	Status
RAM detection	Correct total memory capacity shown in BIOS/OS	Pass / To verify in lab
Storage detection	Correct drive model/capacity visible in BIOS and OS	Pass / To verify in lab

Task 4 - Full Desktop PC Assembly

Objective: Assemble a complete desktop PC in a correct sequence while following ESD and safety procedures, then confirm a successful POST.

Recommended Installation Sequence



1. Prepare a clean workspace, gather tools, disconnect mains power and apply ESD precautions.

2. Inspect the case and install/confirm the correct motherboard standoffs for the motherboard form factor.

3. Install the CPU into the motherboard socket by aligning the socket indicators. Never force the CPU.

4. Apply thermal compound if the cooler does not already have pre-applied material, then install the CPU cooler and connect CPU_FAN.

5. Install RAM in the motherboard-recommended slots for dual-channel operation.

6. Install the M.2 NVMe SSD and secure it to the correct standoff.

7. Install the motherboard I/O shield if it is separate, then place the motherboard into the case and fasten it without overtightening.

8. Install the PSU with its fan facing the correct ventilation direction and route the main cables.

9. Connect 24-pin ATX motherboard power and 8-pin CPU EPS power.

10. Install the GPU into the primary PCIe x16 slot if required; secure it to the case and attach GPU power connectors.

11. Install any SATA drives and connect SATA data plus SATA power.

12. Connect front-panel power/reset/LED leads, front USB headers, front audio and case-fan headers.

13. Inspect the complete build for loose screws, unconnected power, blocked fans and incorrectly routed cables.

14. Connect monitor, keyboard and power. Switch on the PSU and press the case power button.

15. Confirm POST, enter BIOS/UEFI, verify CPU/RAM/storage and check CPU temperature.

16. Configure boot order and install or boot the operating system as required.

Assembly Documentation Record

Step	Evidence / Observation
CPU installed	CPU aligned correctly; socket retention mechanism locked.
Cooler installed	Cooler firmly mounted; CPU_FAN connected.
RAM installed	Modules seated evenly; latches locked.
NVMe installed	Drive fully inserted and secured.
Motherboard installed	Correct standoffs used; screws secure.
PSU connected	24-pin ATX and CPU EPS connected; GPU/SATA power as needed.
GPU installed	Card fully seated and secured; external power connected if required.
Front-panel connected	Power switch, LEDs, USB and audio connected to correct headers.
Cable/airflow check	No loose cables touching fans; airflow paths clear.
POST result	Expected result: POST screen/BIOS available and hardware detected.

Photo/diagram evidence: In a physical assessment, insert labeled photos after the relevant assembly steps. If photos are not available in the written report, the motherboard diagram in Task 2 and the step-by-step evidence table above provide a structured record of the installation sequence.

Scenario: ABC Solutions is setting up a new computer lab. Some newly assembled systems fail to start, while others display hardware-detection errors. The IT team suspects incorrect installation or compatibility problems. The following report combines component selection, installation verification and troubleshooting.

1. First Three Hardware Components to Check

The first checks should target components that can prevent POST entirely and that are commonly disturbed during assembly.

Priority	Component	Why Check It First	What to Check
1	PSU / power connections	Without correct power, the motherboard and CPU cannot initialize. A missing CPU EPS cable can cause fans to spin without a successful POST.	PSU switch, AC cable, 24-pin ATX, CPU EPS, GPU power, front-panel power switch connection.
2	RAM	Poorly seated or incompatible RAM is a very common cause of no-POST conditions and memory diagnostic LEDs.	Module fully seated, correct slot, compatible DDR generation, test one module at a time.
3	CPU / motherboard relationship	The CPU must be correctly installed and supported by the motherboard socket, chipset and BIOS.	Socket match, bent pins/contacts, CPU seating, cooler pressure, BIOS support.

Storage is also important, but a missing storage drive usually does not prevent POST. A system can often enter BIOS successfully even with no boot drive installed. Therefore storage is normally checked after basic POST-critical hardware unless the error message specifically points to the drive.

2. Motherboard-CPU Compatibility and System Functionality

The CPU depends on the motherboard for the correct physical socket, chipset support, voltage regulation and firmware initialization. In the representative workstation, an Intel Core i5-12400 uses the LGA1700 socket, so the motherboard must also provide an LGA1700 socket. The board's BIOS must contain microcode/support for the processor. If the socket does not match, the CPU cannot be installed. If the socket matches but firmware support is missing, the machine may power on but fail to POST or display no video.

For ABC Solutions, a sensible deployment control is to standardize on an approved motherboard/CPU combination and record the required BIOS version before building multiple systems. This reduces the risk of several computers being assembled with the same compatibility fault.

3. Verifying RAM and Storage Installation

RAM verification:

- Check that every DIMM is inserted in the correct orientation and fully seated.
- Check that retention latches are locked and the modules are installed in recommended channel slots.
- At POST/BIOS, verify the total installed memory capacity and, where shown, memory speed/channel information.
- If capacity is incorrect, power off and test modules individually.

Storage verification:

- For NVMe, check that the drive is fully inserted in the correct M.2 slot and secured at the correct standoff length.
- For SATA, check both data and power cables and confirm the motherboard SATA port is enabled.
- In BIOS/UEFI, confirm that the drive model/capacity appears in storage or boot information.

- After OS boot, confirm the drive appears in Disk Management/Device Manager and that the expected partitions are present.

Expected BIOS/POST confirmation: The POST process should complete without memory/storage-related error indicators. BIOS should report the correct total RAM and should list the installed SSD. If the SSD contains the operating system, it should also be available as a boot option.

4. Hardware-Installation Checklist to Prevent Future Failures

- ☐ Confirm motherboard, CPU, RAM and storage compatibility from manufacturer documentation before opening components.
- ☐ Use ESD precautions and disconnect AC power before installing or reseating components.
- ☐ Install the CPU without force and verify the socket alignment indicator before closing the retention mechanism.
- ☐ Install the CPU cooler correctly and connect CPU_FAN.
- ☐ Install RAM in the recommended slots and press until the latches fully engage.
- ☐ Install NVMe/SATA storage securely and verify all required cables.
- ☐ Connect both the 24-pin ATX and CPU EPS power connectors; verify GPU power where needed.
- ☐ Inspect motherboard standoffs and remove any unnecessary standoff that could short the board.
- ☐ Perform a pre-power visual inspection for loose screws, blocked fans and disconnected headers.
- ☐ On first boot, record POST result and verify CPU, RAM, storage and temperatures in BIOS before OS deployment.

5. Technical Justification for SSDs in the ABC Solutions Lab

SSDs are more suitable than HDDs for modern office lab workstations because the lab will repeatedly boot Windows, log in users, launch productivity applications, open documents, install updates and access many small files. These operations benefit strongly from low access latency and high random I/O performance. HDDs use moving heads and rotating platters, so every non-sequential request may require mechanical positioning. SSDs access flash memory electronically and therefore respond much faster.

- Faster boot and login means less time waiting at the start of classes or shifts.
- Applications such as browsers, office suites and collaboration tools open faster.
- Updates and software installation complete more quickly.
- No moving parts means quieter operation and better resistance to vibration or accidental movement.
- Lower power use and heat output are useful when many workstations operate in one lab.
- NVMe SSDs also avoid SATA interface limits and provide strong performance for future workloads.

Therefore, a 500 GB or 1 TB SSD per workstation is a practical standard for ABC Solutions, with centralized/network storage or backup systems used for large shared data where appropriate.

Item 1 - Exact Prompt Given to the AI Tool

Prompt: "Create a step-by-step troubleshooting checklist for a junior technician diagnosing a desktop PC that fails to boot immediately after assembly. Put the checks in priority order. Cover at minimum the PSU, motherboard, RAM, and storage. Use a clear on-site checklist or simple decision flow. Apply it to the ABC Solutions computer-lab scenario and identify which step would most likely catch an assembly-related boot or hardware-detection issue."

Item 2A - AI Original Checklist

The following is the original AI-produced checklist before manual review:

☐ **Step 1:** Confirm the monitor is powered on and connected to the correct video output.

☐ **Step 2:** Check the PSU rear switch and AC cable, then press the case power button.

☐ **Step 3:** Open the case and check the 24-pin motherboard power connector and CPU 8-pin power connector.

☐ **Step 4:** Check whether the storage drive is connected and appears in BIOS.

☐ **Step 5:** Reseat the RAM and try one module at a time.

☐ **Step 6:** Reseat the GPU and verify any PCIe power cables.

☐ **Step 7:** Check motherboard diagnostic LEDs or beep codes.

☐ **Step 8:** If the problem remains, check CPU installation and motherboard-CPU compatibility.

☐ **Step 9:** Use an antistatic wrist strap before continuing to swap components.

☐ **Step 10:** For ABC Solutions, the RAM reseating step would likely catch a system that powers on but does not POST because a DIMM was not fully inserted.

Item 2B - Test & Debug Without AI: Problems Found

During manual review, I identified two weaknesses in the AI checklist:

- ESD protection is mentioned too late. The checklist tells the technician to open the case and touch internal connectors before requiring antistatic precautions. This is unsafe practice.
- Storage is checked before RAM even though a missing storage drive normally does not prevent POST. For a no-boot/no-POST condition immediately after assembly, power and RAM are higher-priority checks.

Item 2C - Corrected Technician Checklist

☐ **Step 1:** Safety first: Shut down if necessary, switch the PSU off, unplug AC power before opening the case, discharge residual power, and apply ESD precautions before touching internal hardware.

☐ **Step 2:** External basics: Confirm monitor power/input, display cable and correct video output. Confirm the AC cable and PSU switch.

☐ **Step 3:** No power at all? Check front-panel power-switch wiring, 24-pin ATX power and CPU EPS power. Confirm the PSU is suitable and known-good if possible.

☐ **Step 4:** Power present but no POST? Check motherboard diagnostic LEDs/beep codes to identify CPU, DRAM, VGA or BOOT stage.

☐ **Step 5:** RAM check: With power disconnected, reseat RAM. Verify correct DDR type and recommended slot. Test one module at a time.

☐ **Step 6:** Graphics check: Reseat the GPU and verify its power cable. If supported, test integrated graphics to isolate the discrete GPU.

☐ **Step 7:** CPU/motherboard check: Verify CPU socket compatibility, BIOS support, correct CPU seating, socket contacts/pins and cooler installation.

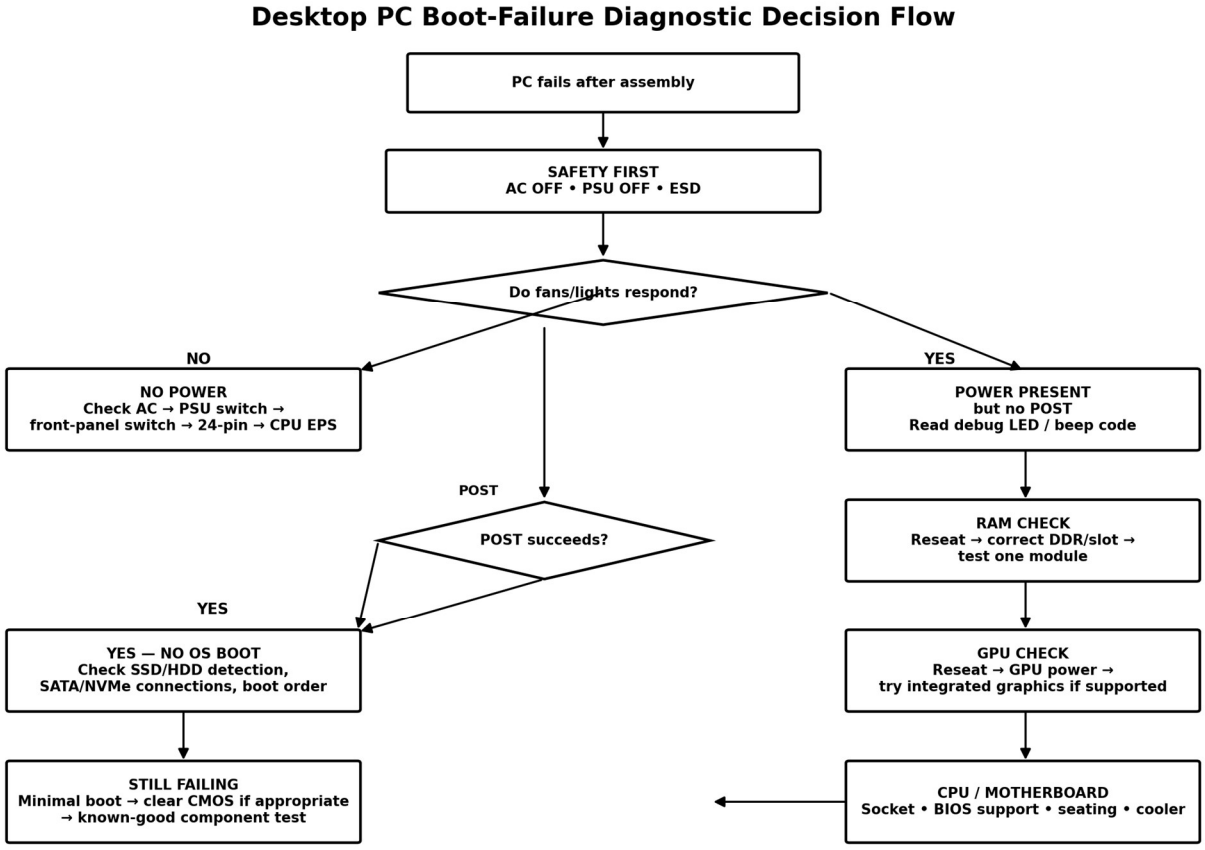
☐ **Step 8:** Minimal boot: Disconnect nonessential storage and peripherals. Attempt POST with motherboard, CPU/cooler, one RAM module and required graphics only.

☐ **Step 9:** POST succeeds but operating system will not boot? Now check storage. Confirm NVMe seating or SATA data/power, then verify that BIOS detects the drive and that boot order is correct.

☐ **Step 10:** If hardware is still not detected, clear CMOS where appropriate and retest with known-good compatible RAM/PSU/GPU one component at a time.

☐ **Step 11:** ABC Solutions application: A machine that powers on but shows a memory/DRAM error would be caught at Step 4 and Step 5. A machine that POSTs but reports no boot device would be caught at Step 9.

Simple Decision Flow




```
PC fails immediately after assembly
  |
  +-- No fans/lights? --> Check AC, PSU switch, front-panel switch, 24-pin ATX, CPU EPS,
```

```
PSU
  |
  +-- Fans/lights but no POST? --> Read debug LED/beep --> Reseat/test RAM --> Check GPU --
> Check CPU/BIOS
  |
  +-- POST succeeds but no OS? --> Check SSD/HDD detection --> Check boot order --> Check
OS/drive
  |
  +-- Still failing? --> Minimal boot --> Clear CMOS if appropriate --> Known-good
component substitution
```

Item 3 - 3-4 Line Reflection Note

I moved ESD and power-isolation precautions to the beginning because a technician should not touch internal components before controlling static electricity and mains power. I also moved the storage check after the POST-critical checks because a missing drive normally does not stop the motherboard from completing POST. The corrected order checks power, motherboard diagnostics and RAM first, which is safer and more efficient for a newly assembled PC. The revised flow also separates a no-POST fault from a POST-successful but no-boot-device fault.

Appendix A - Representative Workstation Specification

This specification is used consistently throughout the model answers. It can be replaced with the exact lab equipment details if required.

Part	Representative Specification	Compatibility Reason
CPU	Intel Core i5-12400, LGA1700	Matches LGA1700 motherboard and is suitable for office/productivity workloads
Motherboard	B660 micro-ATX, LGA1700, DDR4, M.2 NVMe	Supports CPU socket, DDR4 memory and NVMe storage
RAM	16 GB (2 x 8 GB) DDR4	Matches DDR4 DIMM slots and supports dual-channel operation
Storage	1 TB M.2 2280 NVMe SSD	Matches motherboard M.2 NVMe slot
GPU	Integrated graphics or entry discrete GPU	Integrated graphics is sufficient for office use; discrete option can support mild design work
PSU	Quality 550 W unit with 24-pin ATX and CPU EPS	Provides appropriate capacity and required connectors
Cooling	CPU heatsink/fan plus case intake and exhaust	Maintains safe temperatures under sustained load
Case	Micro-ATX compatible case	Physically supports the motherboard form factor

- ☐ CPU model and motherboard socket/chipset compatibility confirmed.
- ☐ BIOS version supports the selected CPU.
- ☐ RAM generation, capacity and slots verified.
- ☐ CPU cooler installed and CPU_FAN connected.
- ☐ NVMe/SATA storage installed and secured.
- ☐ Motherboard mounted on correct standoffs only.
- ☐ 24-pin ATX and CPU EPS power connected.
- ☐ GPU fully seated and powered if required.
- ☐ Front-panel switch/LED headers connected correctly.
- ☐ No loose screws or tools inside the case.
- ☐ Cables clear of fans and airflow paths.
- ☐ POST completed successfully.
- ☐ BIOS detects correct CPU, RAM and storage.
- ☐ CPU temperature normal in BIOS.
- ☐ Correct boot drive selected.
- ☐ Operating system boots successfully.
- ☐ Device Manager/system information checked for missing hardware or drivers.
- ☐ Workstation ready for client handover.